A9. D8 and Theorem 6a, addition to the commentary: the anharmonic stratification, and the one collapse the double slit admits

Insertion points: the taxonomy paragraph following Theorem 6a (split-harmonic, bi-harmonic, …) and the harmonic-conjugate remark of Theorem 7. Companion to the Overflow studies (the-ceiling-of-the-tower, the-equianharmonic-collapse_v2, modular-corner). Conventions and the calibrated rig (screen z = 0, ℓ_1 vertical at Zv = 150, ℓ_2 horizontal at Zh = 250, D = Zh − Zv) are those of the main document. Everything below is verified to machine precision in `check_orbitcollapse.js`.

The classical stratification, recalled

A cross-ratio is one number attached to four points only after an ordering is chosen. Reordering acts through S_4, whose Klein four-group acts trivially, so the effective group is S_3 and the generic orbit has six values:

λ, $1/\lambda$, $1 - \lambda$, $1/(1 - \lambda)$, $\lambda/(\lambda - 1)$, $(\lambda - 1)/\lambda$.

The three involutions of S_3 act on λ-space as the Möbius involutions $\lambda \mapsto 1/\lambda$, $\lambda \mapsto 1 - \lambda$, $\lambda \mapsto \lambda/(\lambda - 1)$, with fixed-point pairs {−1, 1}, {½, ∞}, {2, 0} respectively — each involution fixing one *harmonic* value and one *degenerate* one. Over ℝ this stratifies the non-degenerate values into exactly two strata: the generic (anharmonic) stratum, orbit size 6, and the harmonic orbit {−1, 2, ½}, size 3. The equianharmonic stratum — fixed points of the 3-cycles, $\lambda^2 - \lambda + 1 = 0$, where the orbit would shrink to 2 — is empty over the reals: the quadratic has discriminant −3 and real minimum 3/4 (verified: min 0.750000). The complete invariant of the orbit is the j-invariant $j(\lambda) = 256(\lambda^2 - \lambda + 1)^3 / (\lambda^2(\lambda - 1)^2)$, constant on the orbit to relative 10^{-15} in the check, with j = 1728 on the harmonic stratum (verified exactly) and the empty equianharmonic stratum sitting at j = 0.

The diagonal action, and where the collapse survives

In double-slit geometry the invariant of four pairwise non-parallel rays is not one number but the pair $[\![r_1, r_2; r_3, r_4]\!] = (\lambda, \mu)$ of D8 — one cross-ratio per slit, a single cross-ratio valued in ℝ ⊕ ℝ. Reordering the *rays* reorders both foot-quadruples at once, so S_3 acts **diagonally**: $\sigma\cdot(\lambda, \mu) = (\sigma\lambda, \sigma\mu)$. The orbit of a pair therefore has size $6/|\mathrm{Stab}(\lambda) \cap \mathrm{Stab}(\mu)|$, and the intersection is the whole story:

Claim (orbit collapse is a chain phenomenon). For a non-degenerate pair (λ, μ) over $\mathbb{R} \oplus \mathbb{R}$, the diagonal orbit has size 6 unless $\lambda = \mu \in \{-1, 2, \tfrac{1}{2}\}$, where it has size 3. In particular the classical six-to-three harmonic collapse survives in double-slit geometry **only on the real diagonal** — by Theorem 6b, only for quadruples that are the image of one straight line — and only at harmonic position on that line.

The reason is that each harmonic value is fixed by a *different* involution, so two harmonic coordinates share a stabilizer only when they are the same value. The two configurations one would expect to collapse, and which do not, are worth recording as negative controls. A **split-harmonic** quadruple ($\lambda = -1$, μ generic; constructed on the live rig with $\mu = 1.2$) keeps a full six-element diagonal orbit. So does a **mismatched bi-harmonic** quadruple ($\lambda = -1$, $\mu = 2$: harmonic on both slits, by different involutions) — six elements again, verified. Only the co-chained harmonic quadruple collapses: four scene points on one line, the fourth solved to the harmonic conjugate, give $\lambda = \mu = -1.000000000$ on both slits and a diagonal orbit of exactly three pairs, $\{(-1, -1), (2, 2), (\tfrac{1}{2}, \tfrac{1}{2})\}$. The taxonomy of Theorem 6a thus carries an orbit count: split-harmonic and mismatched bi-harmonic are six-fold strata that merely *contain* harmonic values; the genuinely collapsed stratum is bi-harmonic-on-the-diagonal, and membership in it certifies both harmonicity and co-chaining at once.

A refinement of the same point: the pair of j-invariants $(j(\lambda), j(\mu))$ is a diagonal invariant but not a complete one. The pairs (λ, λ) and $(\lambda, 1/\lambda)$ have identical j-pairs and lie in different diagonal orbits (verified at $\lambda = 3.7$) — the orbit remembers chain membership where the j-pair forgets it. The information the j-pair discards is exactly the skew defect of Theorem 6b.

The collapse the double slit cannot have

Over $\mathbb{R} \oplus \mathbb{R}$ the equianharmonic stratum stays empty: $\lambda^2 - \lambda + 1 = 0$ would have to hold in each component separately, and it holds in neither. So of the Benz trinity, the equianharmonic collapse — six values to two, $j = 0$ — is the property of the *elliptic* member, $\mathbb{C}$, where the Overflow studies stage it; the split member can be harmonic but can never be equianharmonic. Six-to-three is the deepest collapse this geometry admits, and it is reserved for chains.

The physical harmonic address

The harmonic stratum is not only reachable; the rig points at it. On the ray through a scene point P, the four canonical points of Theorem 7 sit at

$$(F_1, F_2; S, P) = -1 \iff z = 2 \cdot Zv \cdot Zh/(Zv + Zh) = 187.5,$$

the harmonic mean of the slit depths — recovered blind by root-finding to nine decimals in the check. At that depth the two magnifications have equal magnitude and opposite parity:

$AR(z^*) = Zh(z - Zv)/(Zv(z - Zh)) = -1$ exactly, confirmed both from the formula and by direct projection of a small square ($sy/sx = -1.000000000$). Consequently the six reorderings of the canonical quadruple yield only three distinct readings, $\{-1, ½, 2\}$, at $z = z^*$, and six everywhere else (six at $z = 200$, verified). This is the same surface that appears in the dual-rig development as the shape-horopter, the zero-disparity locus where the two channels fuse. One invariant, three appearances: the orbit collapse of the reordering group, the parity balance of the magnifications, and the fusion depth of the doubled camera are a single harmonic fact read in three charts.

Verification summary

Addendum	Script	Key figures
A9	check_orbitcollapse.js	group closure exact; harmonic orbit $\{-1, ½, 2\}$; j constant to rel 10^{-15}, $j(-1) = 1728$; co-chained $\lambda = \mu$ to 4×10^{-15}; co-chained harmonic orbit 3; split-harmonic and mismatched bi-harmonic orbits 6; $z^* = 187.500000000$ by root-find; $AR(z^*) = -1$ by direct projection; equianharmonic minimum 3/4

Script runs under node with no dependencies, in the discipline of the A1–A8 checks; 22 assertions, all passing against the live rig constants.